Team 12 Case Study

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Introduction

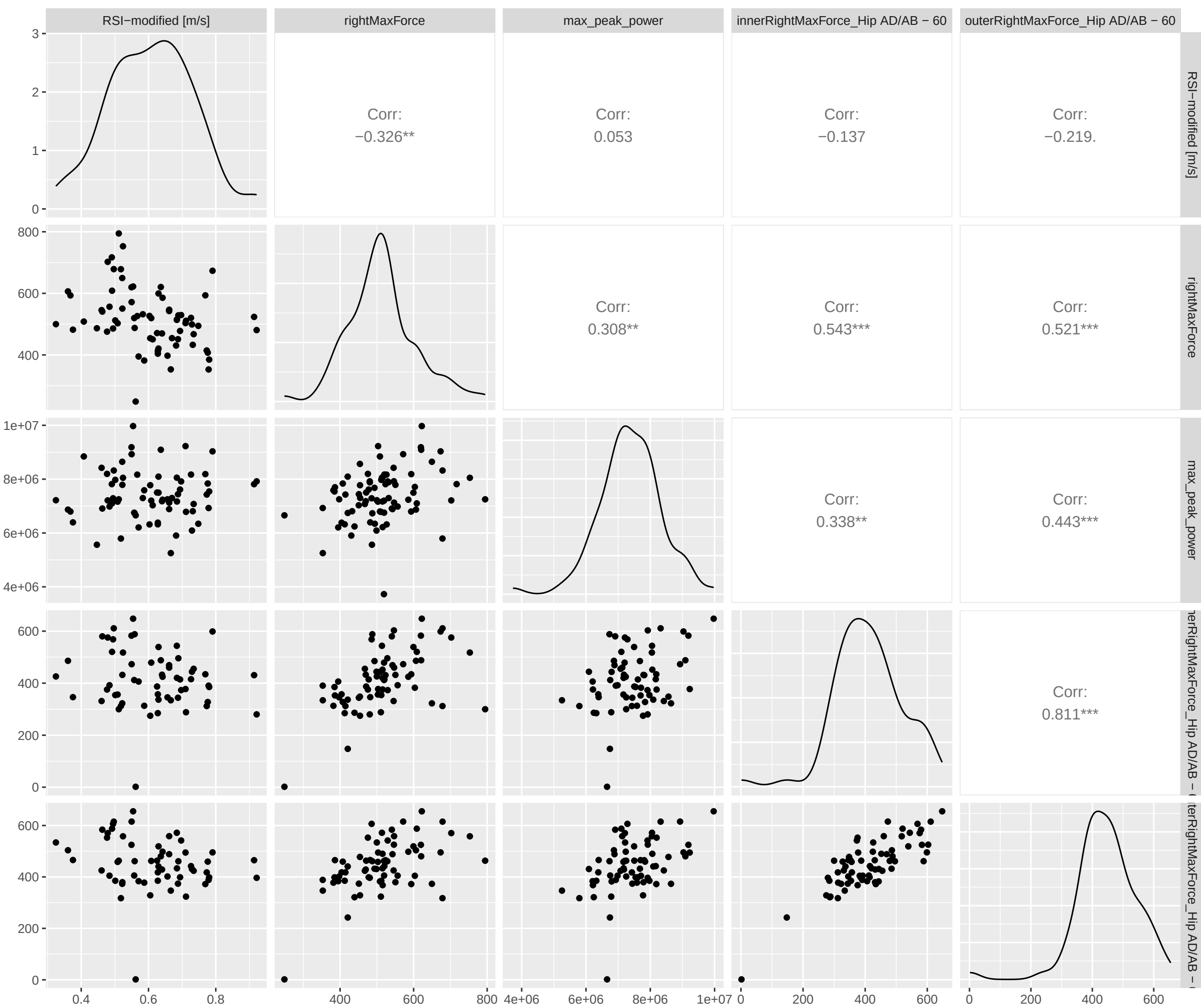
Duke Football is interested in using biomechanics data to improve performance and prevent injury in athletes. Using four instruments, they have collected physiometric data from athletes on the football team over the course of roughly a year. Athletes used each of these intruments multiple times over the course of that period.

The four instruments used to collect biomechanical data are `forcedeck`, `forceframe`, `nordbord`, and `perch`. The `forcedeck` instrument measures the power, acceleration, and other mechanics generated from the athlete's lower body when the athlete jumps straight up while their hands are on their hips. The `forceframe` instrument asseses the isometric strength of an athlete's different joints, especially hips, by measuring the force generated from the athlete's legs when they push their legs in or out while lying on their back. The `nordbord` instrument calculates hamstring stregth by measuring the force generated from the lower legs when the athlete lowers their body from a kneeling position. The `perch` instrument measures the power and velocity generated by the athlete while performing a squat with weights. Data from the football players were collected from these instruments and provided to the statistical team as four separate datasets.

We focused our analysis on two research questions. First, we investigated whether performance metrics from three of the instruments could be used to find associations with performance on a fourth instrument. Second, we examined whether asymmetries in performance (differences between left and right limbs) recorded on one instrument were associated with performance on another instrument. Understanding these two types of associations can not only improve the physical preparation of athletes and their performance but also identify weaknesses that affect performance in other, seemingly unrelated body parts, and prevent injuries.

The research described above was motivated by understanding that each of these test individually measures a single body part but an athlete needs its full body in peak form in order to perform at the highest level. Therefore, athletes preparation for elite competition can increase in efficiency if there exists body parts which increased performance is associated with increased performance in others. In other words, we intend to find a way to increase overall body performance from a increases in performance in a single or couple body parts. The motivation for investigating asymmetries is very similar. We intend to understand the association between imbalances in one body part and performance in others. An association between these might indicate that in a single body part might be associated to decrease in overall performance and not only in a specific test. This is also important in athletes preparation for high level competition.

Exploratory plots of the data show that the data appear to be right-skewed with significant outliers.



Methodology

Data cleaning

For each of the four instruments, we selected a single variable that is most representative of what each of the instruments intends to assess. For `forcedeck`, we selected `RSI-modified` which stands for Reactive Strength Index and is a measure of explosiveness, neuromuscular readiness and fatigue. In the case of `forceframe`, athletes performed multiple tests with this instrument. Most of the tests were performed by less than 30 athletes. Therefore, we chose to analyze performance in the two tests that were performed the most. These were: `Hip Adduction/Abduction` with a 60 degree supine position (70 athletes) and `Shoulder Internal/External Rotations` (83 athletes) with a custom setup. For the hips tests, we chose the `Inner Right Max Force` because this represents the adduction force, which is related to groin and adductor muscle strength. For the shoulders test, we chose the `Outer Right Max Force` because this is the extension force, which is crucial in football, especially linemen. Then, for the `nordbord`, we selected the `Right Max Force` as a measurement of hamstring strength. Finally, for `perch`, we selected `Max Peak Power` as a measure of explosiveness.

It is important to note the Duke Athletics Sports Science team did not express a preference for a side when metrics where available for both left and right limbs. Therefore, we chose to consistently use the right side metrics in all cases where this applied, based on the assumption that it is the dominant side for the majority of athletes. However, this choice is not expected to have any significant impact on our analysis.

To explore asymmetries, we also chose a metric from each of the four instruments. For `forcedeck`, the data already contained the following left/right asymmetry metrics in percentage of asymmetry: `Landing Impulse Asymmetry`, `Peak Landing Force Asymmetry`, and `Eccentric Deceleration RFD Asymmetry`. We calculated left/right asymmetry percentages for `Inner Max Force` and `Outer Max Force` for both the hips and shoulders tests in `forceframe` and for `Max Force` in `nordbord`. To do so, we used the following formula, which was used the calculate the asymmetry percentages already included in the the `forcedeck` dataset: Left/Right Asymmetry Percentage $= \left|\frac{\text{Left+Right}}{\max(\text{Left,Right})}\right| \times 100$. Perch did not provide any metrics from which asymmetries could be calculated.

To analyze relationships across the four datasets, we decided to average measurements for each athlete for each variable of interest. To ensure that information was not lost, we checked that variances between athletes were generally low for the variables of interest (see Appendix). Variances all appeared to be centered near zero and skewed right, with the exception of `Outer Right Max Force`.

We then combined the datasets using the athlete ID as an identifier. Since body weight was found in the `forcedeck` dataset in kilograms and was also collected in the `perch` dataset in pounds, we combined the two variables by using the kilograms value from `forcedeck` if present and if not, converting the value from `perch` to kilograms. Finally, we scaled all variables so that individual variables would not disproportionately influence the outcome of the model by standardizing each variable to a mean of 0 and variance of 1.

Modeling for performance relationships across instruments

To explore the associations between performance across the different instruments, we fitted a set of 5 linear regression models using the scaled data previously described. We used the following variables of interest in each of the models: `Reactive Strength Index` from `forcedeck`, `Inner Right Max Force` from `forceframe` hip tests, `Outer Right Max Force` from `forceframe` shoulder tests, `Right Max Force` from `nordbord`, and `Max Peak Power` from `perch`. In the 5 models, each of these variables served as the response variable while the other four where included as predictors. Moreover, body weight was included in each of the models to account for variability of athlete weights within a football team. Linear model diagnostics were assessed and can be found in the Appendix.

Modeling for the effect of asymmetries in performance of other instruments

To investigate the associations of left/right asymmetries on performance across the different instruments, we conducted another set of 5 linear regression models. The goal of these models was to evaluate the hypothesis that higher asymmetry percentages have a negative association on performance and to identify which asymmetric body parts have these negative effects. We fitted a model for each of the 5 variables we defined to be representatives of performance in the 4 instruments. These target variables are: `Reactive Strength Index` from `forcedeck`, `Inner Right Max Force` from `forceframe` hip tests, `Outer Right Max Force` from `forceframe` shoulder tests, `Right Max Force` from `nordbord`, and `Max Peak Power` from `perch`. The predictors for these five models were all left/right asymmetry percentages. More specifically, these were: `Landing Impulse Asymmetry`, `Peak Landing Force Asymmetry`, and `Eccentric Deceleration RFD Asymmetry` from `forcedeck`; `Inner Max Force Asymmetry` and `Outer Max Force Asymmetry` for both hips and shoulders tests from `forceframe`; and `Max Force Asymmetry` from `nordbord`. In each of the models we only included predictors that came from a different instrument that the one that measured the target variable. Body weight was included in all models to account for the variability across athletes in within the team. Linear model diagnostics were checked and can be found in the Appendix.

Results

Performance Models

Max Peak Power from perch appeared to have a significant positive association with performance on each of the other instruments. More specifically, Reactive Strength Index was significantly associated (p=0.009) with Max Peak Power. We expect a 0.338 unit increase in Reactive Strength Index for every one unit increase in Max Peak Power, holding constant all other predictors in the model. This indicates that athletes who generated a higher peak power during a squad also showed more explosiveness, neuromuscular readiness and less fatigue during their vertical jumps. The variables included from nordbord and forceframe were not statistically significant.

In the case of force deck, only Shoulder Outer Right Max Force was significantly associated (p=0.018) with Max Peak Power but Hip Inner Right Max Force was not. For every one unit increase in Max Peak Power, we expect a 0.319 unit increase in Shoulder Outer Right Max Force, holding constant all other predictors in the model. This suggests that players who generated higher peak power during a squad also generated more outer force in their shoulders. Variables from the other instruments included in the model were not significant.

Finally, Right Max Force from nordbord was also significantly associated (p=0.045) with Max Peak Power. For every one unit increase in Max Peak Power, we expect a 0.301 unit increase in Max Right Force, e expect a 0.319 unit increase in Shoulder Outer Right Max Force. This means that athletes who generated higher peak power during a squad also generated higher force with their hamstrings. Predictors from other instruments were not statistically significant.

All of these results were evaluated at the 0.05 alpha level.

Table 1: Linear Regression Model for RSI-modified (forcedeck)

Coefficient	Estimate	Std. Error	t-value	p-value
Intercept	-0.029	0.1	-0.287	0.775
Right Max Force (nordbord)	-0.059	0.122	-0.484	0.631
Max Peak Power (perch)	0.338	0.124	2.735	0.009
Inner Right Max Force Hip AD/AB (forceframe)	0.135	0.142	0.952	0.346
Outer Right Max Force Shoulder IR/ER (forceframe)	0.031	0.135	0.23	0.819
Body Weight	-0.843	0.145	-5.824	<0.001

Table 2: Linear Regression Model for Outer Right Max Force Shoulder IR/ER - Custom (forceframe)

Coefficient	Estimate	Std. Error	t-value	p-value
Intercept	-0.053	0.104	-0.507	0.614
RSI modified (forcedeck)	0.034	0.148	0.230	0.819
Right Max Force (nordbord)	-0.228	0.124	-1.841	0.072
Max Peak Power (perch)	0.319	0.131	2.436	0.018
Inner Right Max Force Hip AD/AB (forceframe)	0.185	0.148	1.257	0.215
Body Weight	0.289	0.191	1.512	0.137

Table 3: Linear Regression Model for Right Max Force (nordbord)

Coefficient	Estimate	Std. Error	t-value	p-value
Intercept	0.056	0.115	0.490	0.627
RSI modified (forcedeck)	-0.079	0.163	-0.484	0.631
Outer Right Max Force Shoulder IR/ER (forceframe)	-0.278	0.151	-1.841	0.072
Max Peak Power (perch)	0.301	0.147	2.052	0.045
Inner Right Max Force Hip AD/AB (forceframe)	0.257	0.161	1.590	0.118
Body Weight	0.466	0.206	2.263	0.028

Table 4: Linear Regression Model for Max Peak Power (perch)

Coefficient	Estimate	Std. Error	t-value	p-value
Intercept	0.092	0.106	0.868	0.390
RSI modified (forcedeck)	0.385	0.141	2.735	0.009
Outer Right Max Force Shoulder IR/ER (forceframe)	0.333	0.137	2.436	0.018
Right Max Force (nordbord)	0.258	0.126	2.052	0.045
Inner Right Max Force Hip AD/AB (forceframe)	0.041	0.153	0.265	0.792
Body Weight	0.261	0.197	1.326	0.191

Asymmetry Models

From the five asymmetry percentage models, the only significant association (p=0.045) was between Hip Inner Right Max Force and Max Force Asymmetry Percentage from nordbord. For every one percentage point increase in Max Force Asymmetry Percentage, we expect Hip Inner Right Max Force to decrease by -0.040 units, holding constant all other predictors in the model. This suggest a negative association between weaknesses or imbalances in an athletes hamstring and their capacity to generate force in their hips. This result was evaluated at the 0.05 alpha level.

Table 5: Asymmetry Linear Regression Model for Inner Right Max Force Hip AD/AB - 60 (forceframe)

Coefficient	Estimate	Std. Error	t-value	p-value
Intercept	0.281	0.232	1.212	0.231
Landing Impulse Asymmetry % (forcedeck)	0.005	0.012	0.398	0.693
Peak Landing Force Asymmetry % (forcedeck)	-0.001	0.014	-0.072	0.943
Eccentric Deceleration RFD Asymmetry % (forcedeck)	-0.008	0.007	-1.042	0.303
Max Force Asymmetry % (nordbord)	-0.04	0.019	-2.054	0.045
Shoulders Inner Max Force Asymmetry % (forceframe)	0.013	0.019	0.7	0.487
Shoulders Outer Max Force Asymmetry % (forceframe)	-0.01	0.015	-0.672	0.505
Body Weight	0.512	0.106	4.832	<0.001

Discussion

The first important conclusion from these models is that the `Max Peak Power` metric from the perch exercise is significantly associated with the metrics of interest from the other variables. This is noteworthy as many of the other metrics of interest do not have this association among themselves, but `Max Peak Power` does. This could mean that the perch exercise should be a point of focus for coaches and athletic trainers as it can act as an indicator for strength, power, neuromuscular readiness, and fatigue in the other exercises.

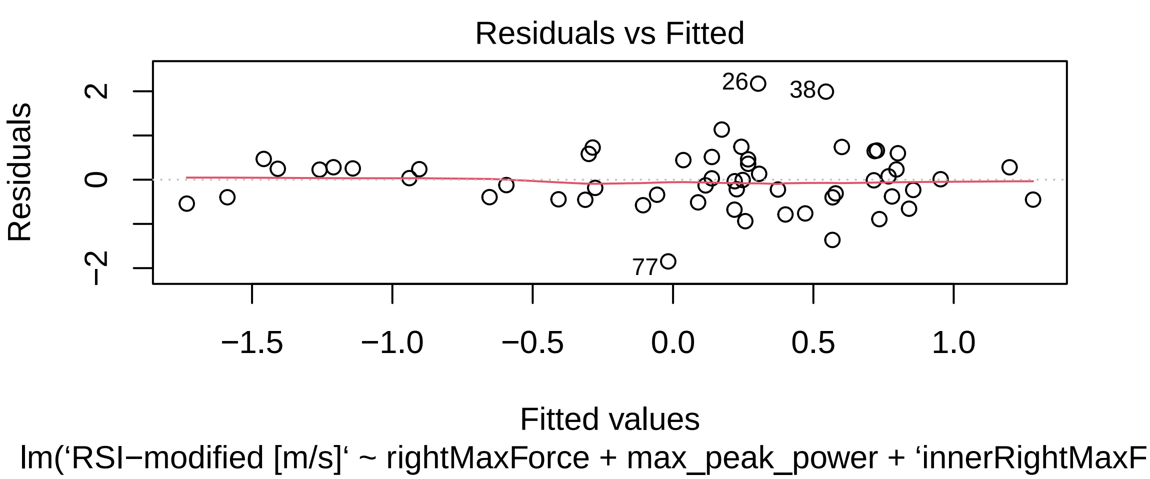
The second main conclusion is from the asymmetry models. These models were fit in an effort to evaluate if strength asymmetries in athletes impact just the relevant exercise or others. Do these imbalances correlate to reduced scores in other metrics? Our models show that those imbalances do not show correlation with strength scores from other parts of the body other than with hamstrings as tested through the nordbord. The fact that hamstring and hip force are negatively correlated may indicate a trade off between these systems of the body. This may also be interesting to Duke football coaches and athletic trainers in terms of injury recovery and prevention and strength training focus.

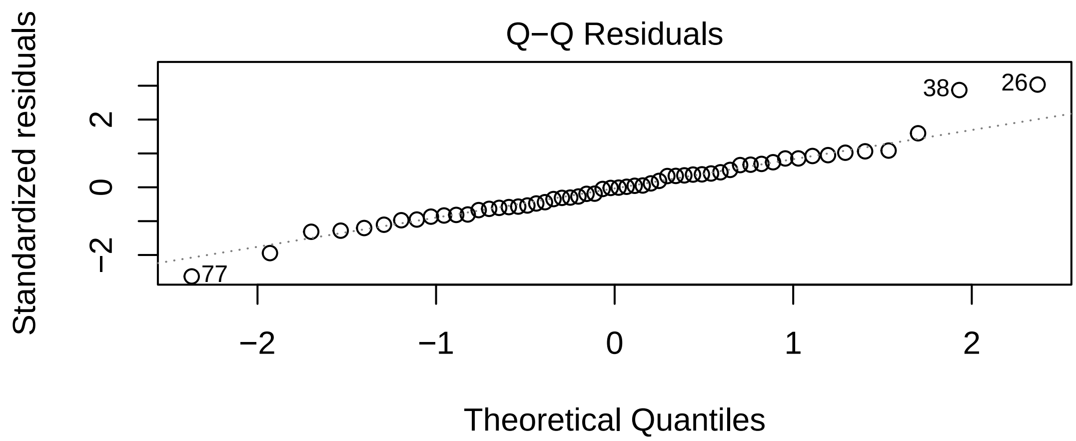
There are of course limits to this analysis, as we are not members of the community that have full access to the data. Without data like player position, injuries, training focus for certain periods of time, we do not have a complete picture. Data errors from mis-reads to missed tests have also complicated the analysis. Moreover, out data and results are only indicative of performance between these instruments but not on-field performance, which is the ultimate goal. Also, our data does not account for the players load differences (startes play and are more tired than players than see less minutes) which might be another factor to consider. Our approach is simplistic but effective, some sort of hierarchical approach could be a future direction for this analysis, relying on athletes within position groups within teams, allowing us to look across schools to find patterns in the general population.

Appendix

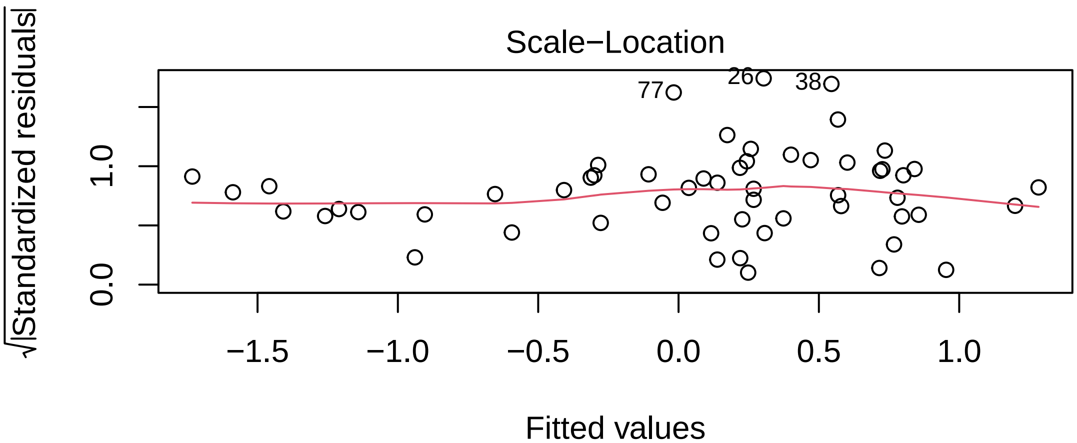
Linear model diagnostics

Linear Regression Model for RSI-modified (forcedeck):

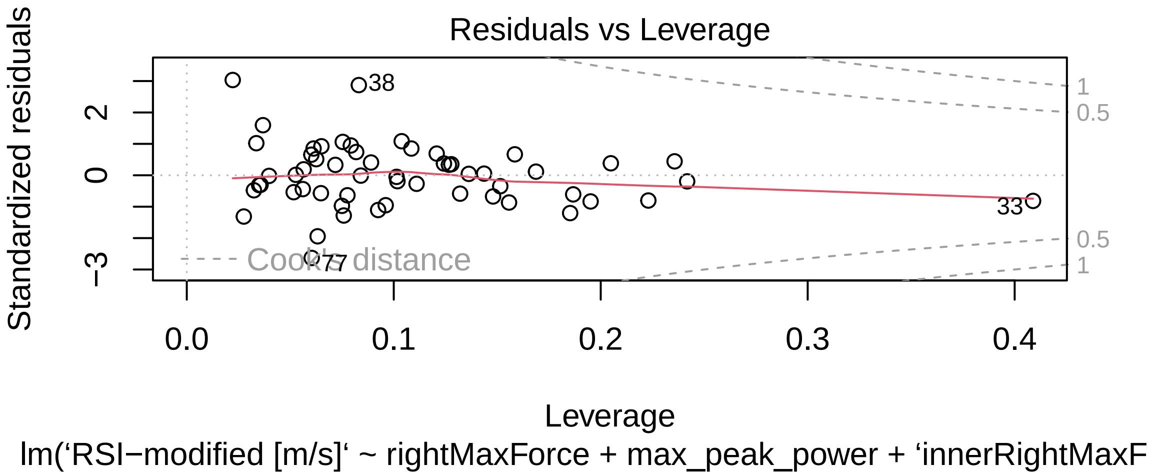




lm(‘RSI−modified [m/s]‘ ~ rightMaxForce + max_peak_power + ‘innerRightMaxF

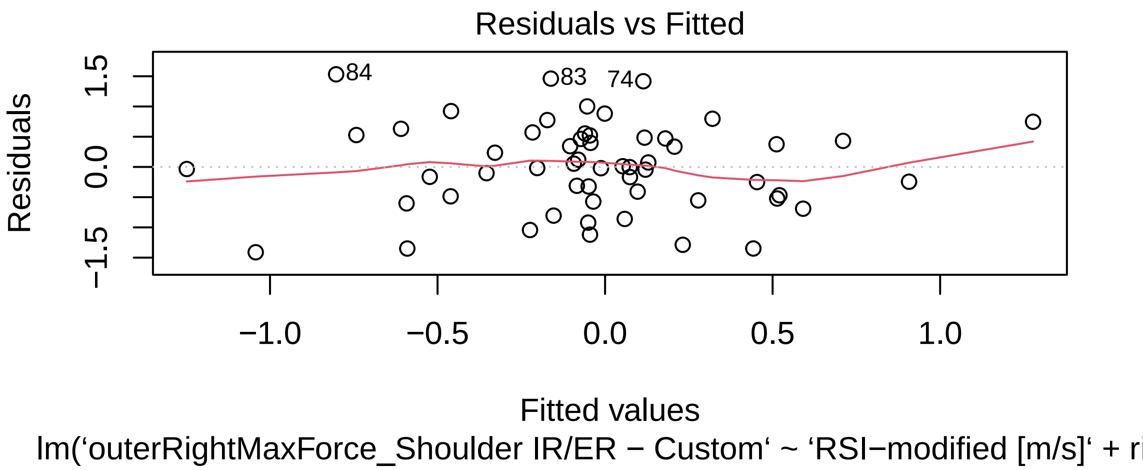


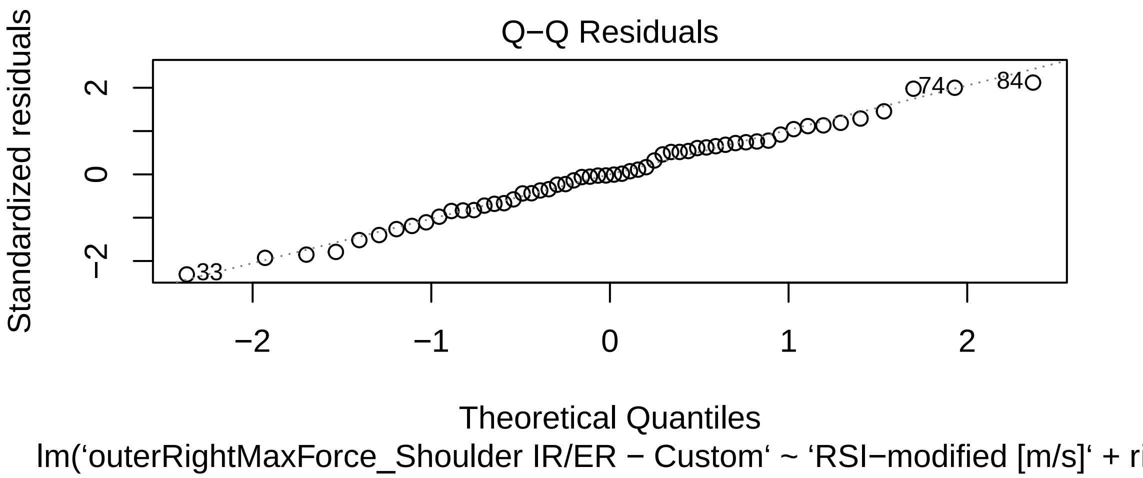
lm(‘RSI−modified [m/s]‘ ~ rightMaxForce + max_peak_power + ‘innerRightMaxF

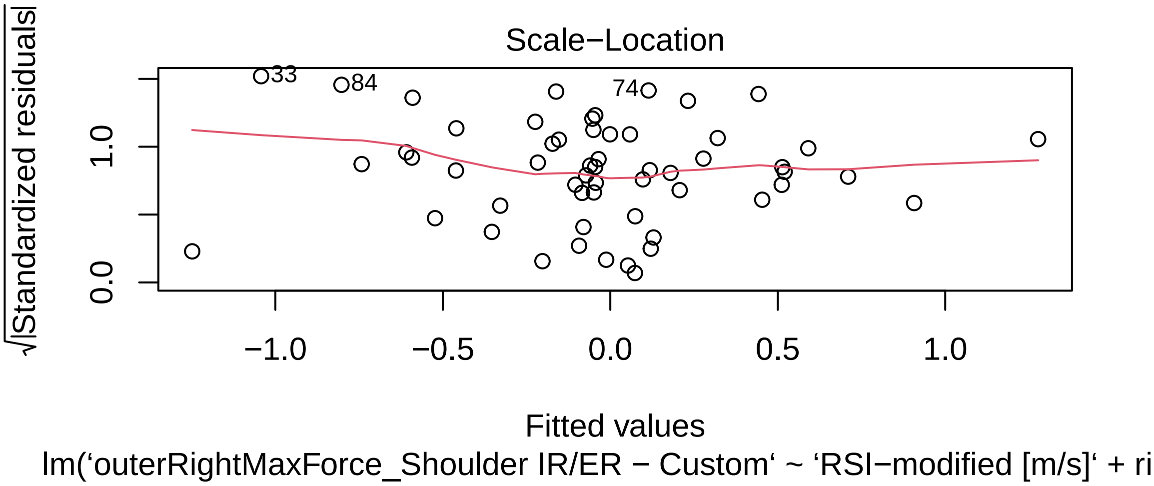


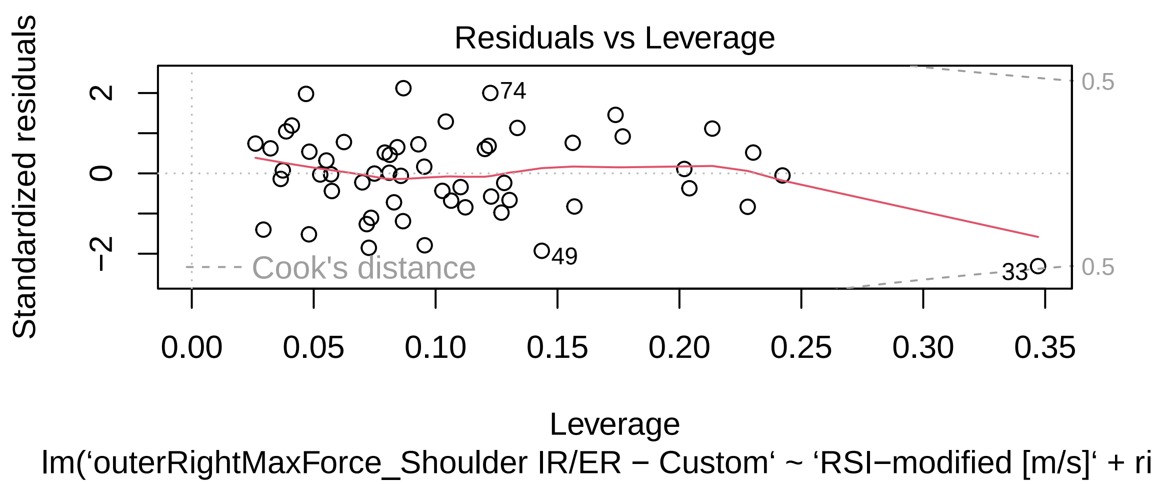
Residuals vs Leverage
Standardized residuals
2
0
−3
38
33
77
1
0.5
0.5
1
Cook's distance
0.0
0.1
0.2
0.3
0.4
Leverage
lm(‘RSI−modified [m/s]‘ ~ rightMaxForce + max_peak_power + ‘innerRightMaxF

Linear Regression Model for Outer Right Max Force Shoulder IR/ER - Custom (forceframe):

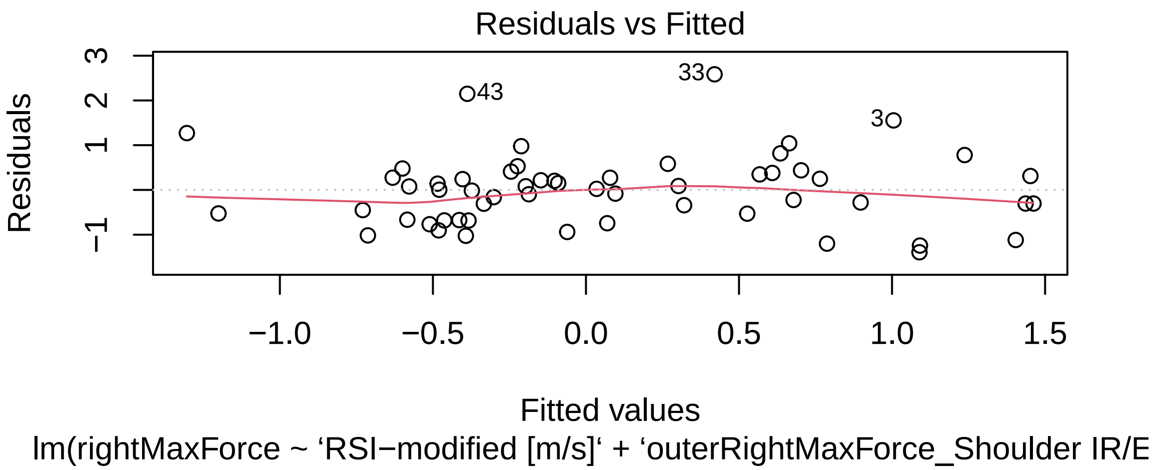


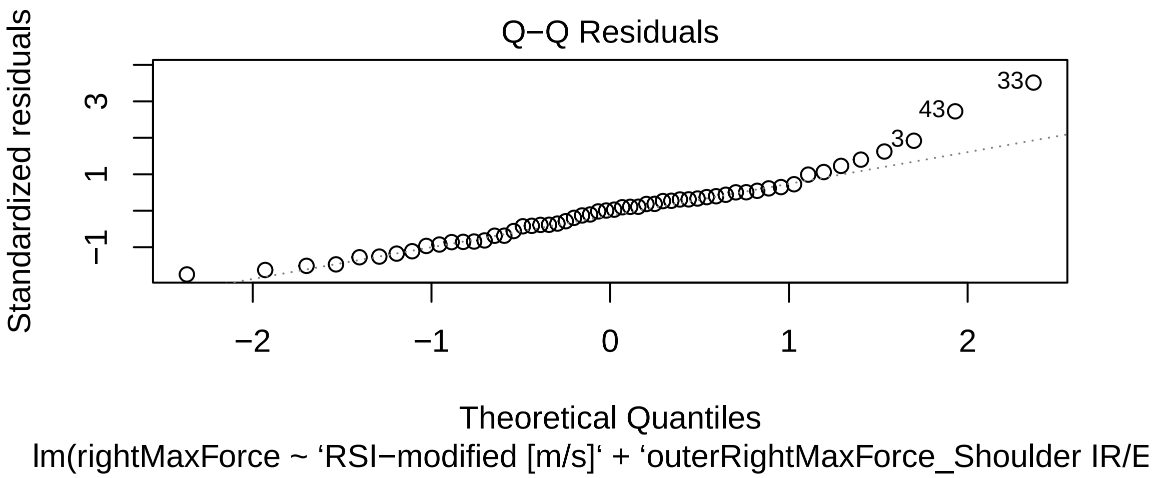


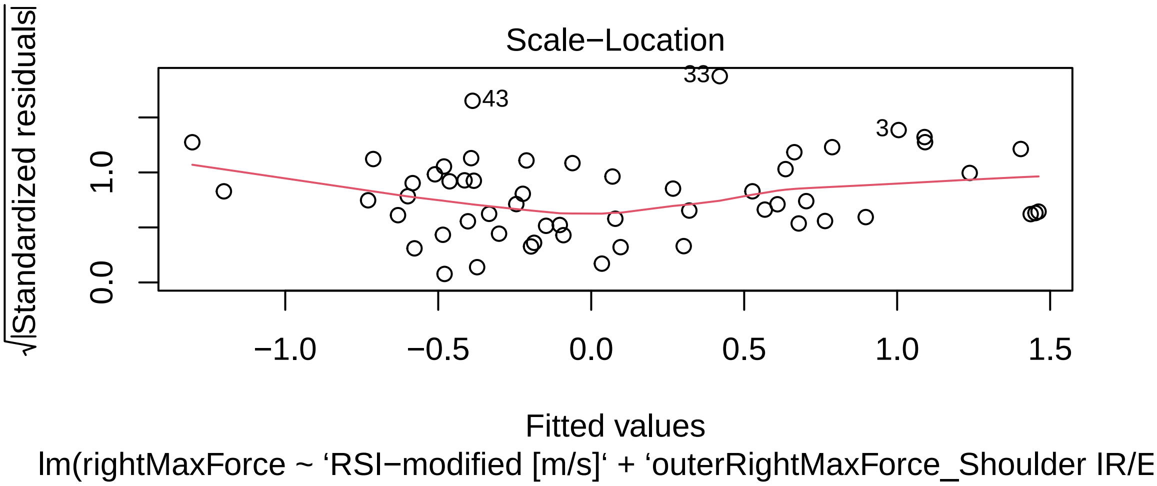

Scale−Location
√|Standardized residuals|
0.0
1.0
−1.0
−0.5
0.0
0.5
1.0
33
84
74
Fitted values
lm(‘outerRightMaxForce_Shoulder IR/ER − Custom‘ ~ ‘RSI−modified [m/s]‘ + ri

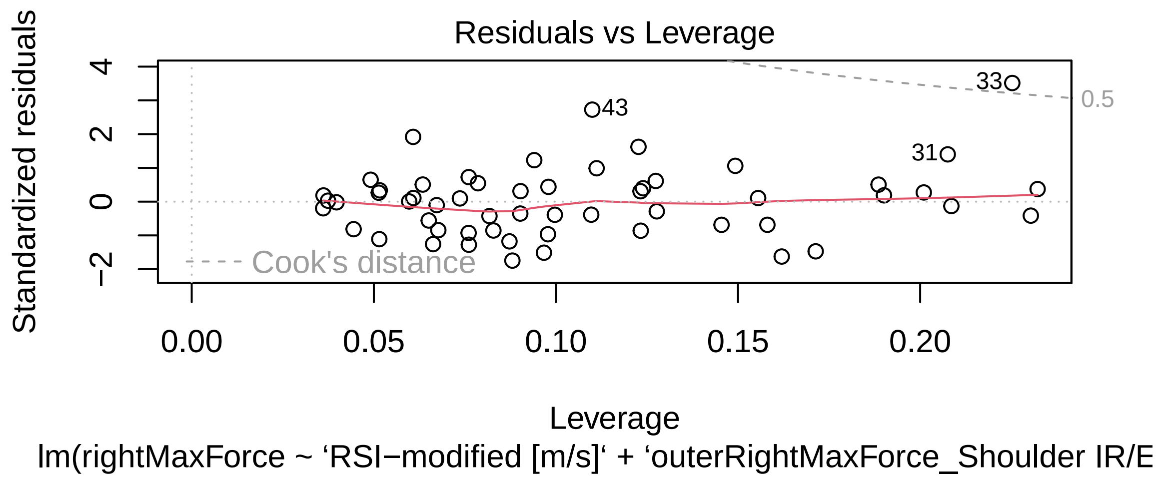

Residuals vs Leverage
Standardized residuals
−2
0
2
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
74
49
33
0.5
0.5
Cook's distance
Leverage
lm(‘outerRightMaxForce_Shoulder IR/ER − Custom‘ ~ ‘RSI−modified [m/s]‘ + ri

Linear Regression Model for Right Max Force (nordbord):

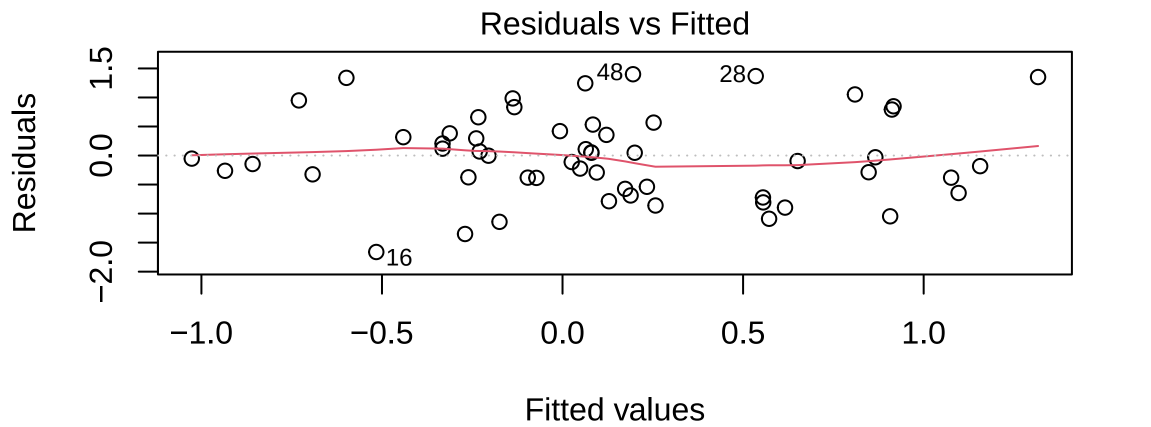




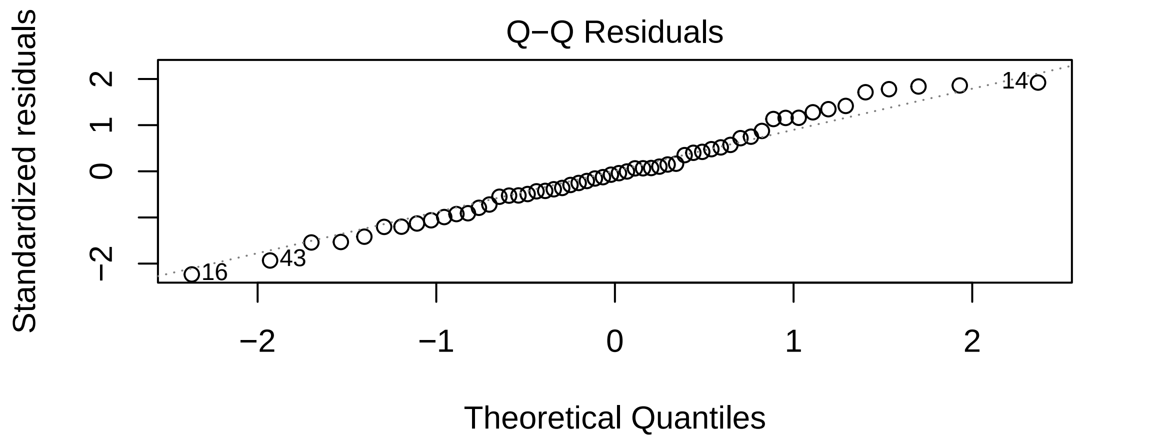

Scale−Location
√|Standardized residuals|
0.0
1.0
−1.0
−0.5
0.0
0.5
1.0
1.5
43
33
3
Fitted values
lm(rightMaxForce ~ ‘RSI−modified [m/s]‘ + ‘outerRightMaxForce_Shoulder IR/E


Residuals vs Leverage
Standardized residuals
−2
0
2
4
0.00
0.05
0.10
0.15
0.20
43
33
31
0.5
Cook's distance
Leverage
lm(rightMaxForce ~ ‘RSI−modified [m/s]‘ + ‘outerRightMaxForce_Shoulder IR/E

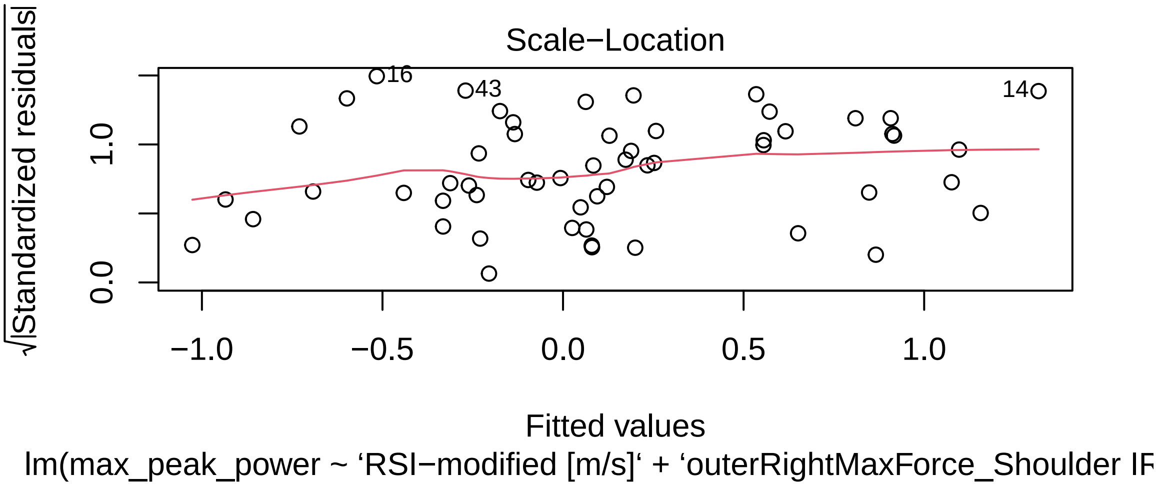
Linear Regression Model for Max Peak Power (perch):

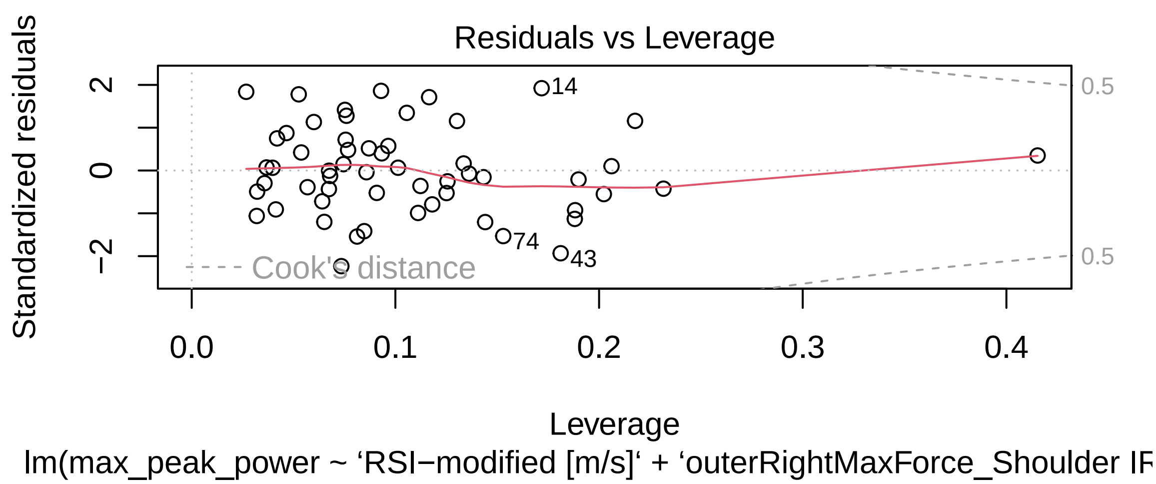


lm(max_peak_power ~ ‘RSI−modified [m/s]‘ + ‘outerRightMaxForce_Shoulder IF

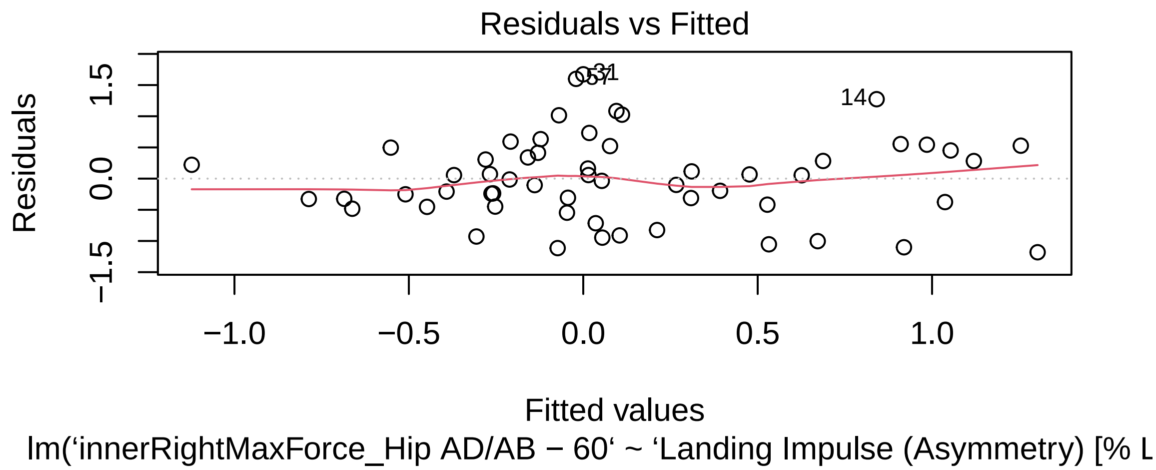


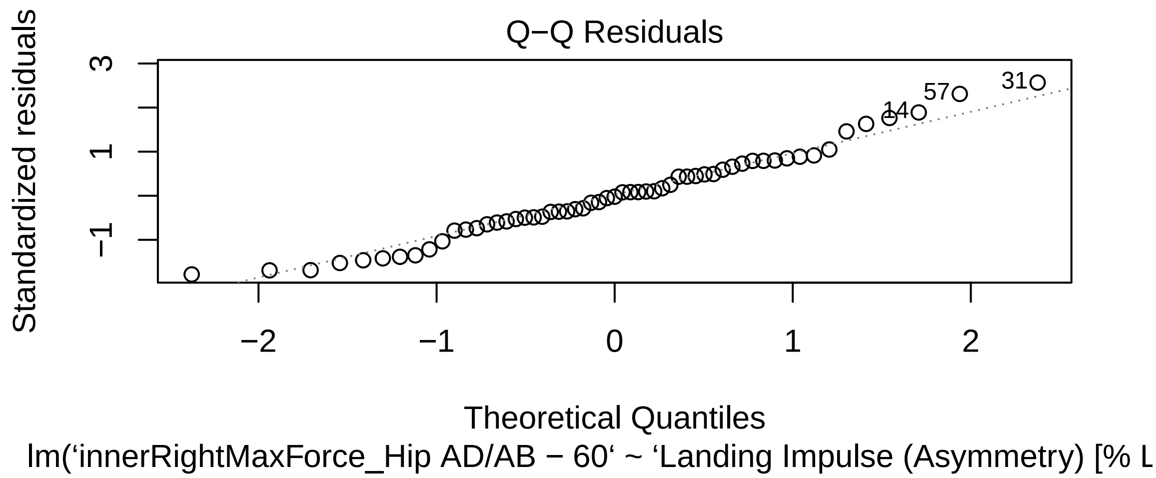
lm(max_peak_power ~ ‘RSI−modified [m/s]‘ + ‘outerRightMaxForce_Shoulder IF

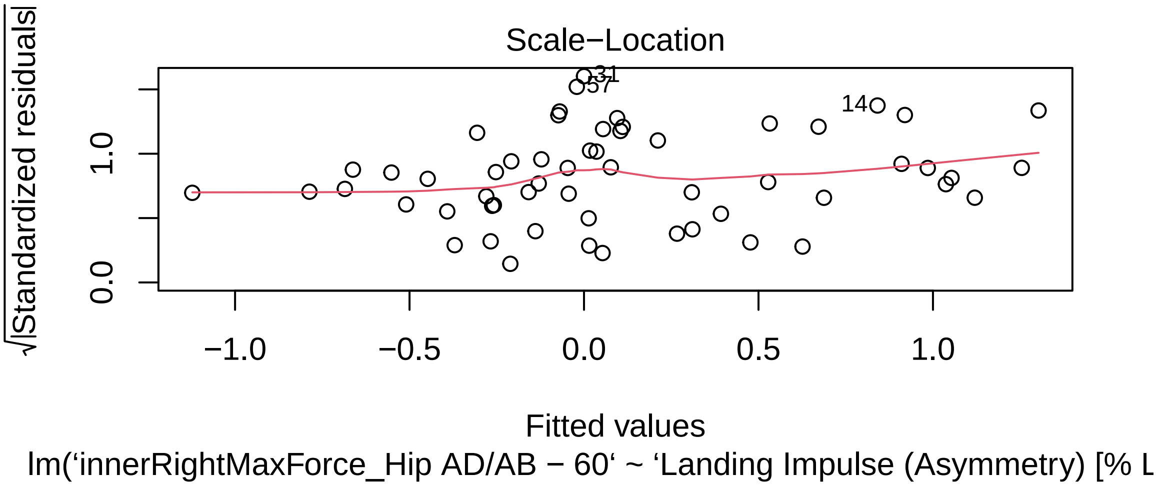

Scale−Location
√|Standardized residuals|
16
43
14
Fitted values
lm(max_peak_power ~ ‘RSI−modified [m/s]‘ + ‘outerRightMaxForce_Shoulder IF


Residuals vs Leverage
Standardized residuals
14
74
43
Cook's distance
0.5
0.5
Leverage
lm(max_peak_power ~ ‘RSI−modified [m/s]‘ + ‘outerRightMaxForce_Shoulder IF

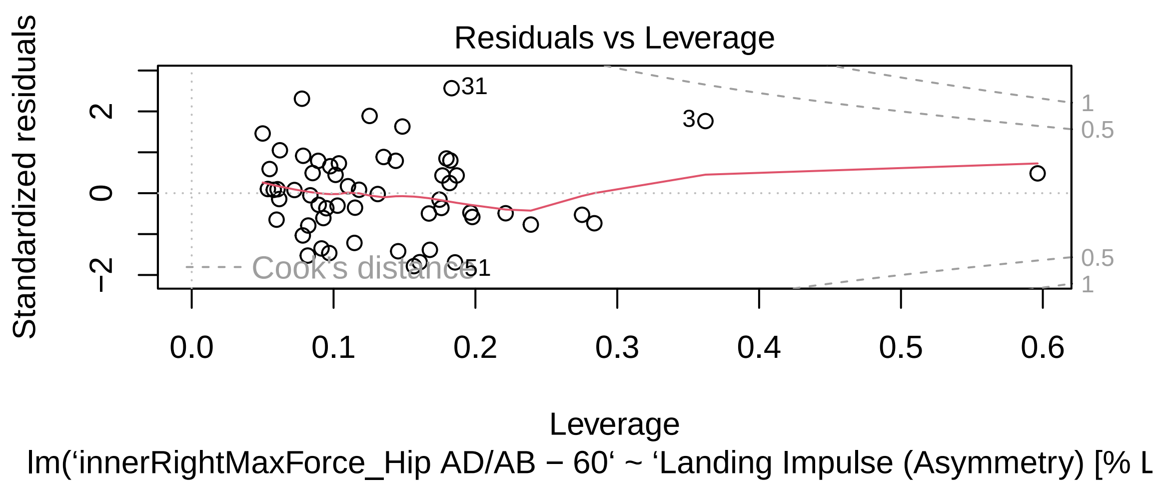
Asymmetry Linear Regression Model for Inner Right Max Force Hip AD/AB - 60 (forceframe):







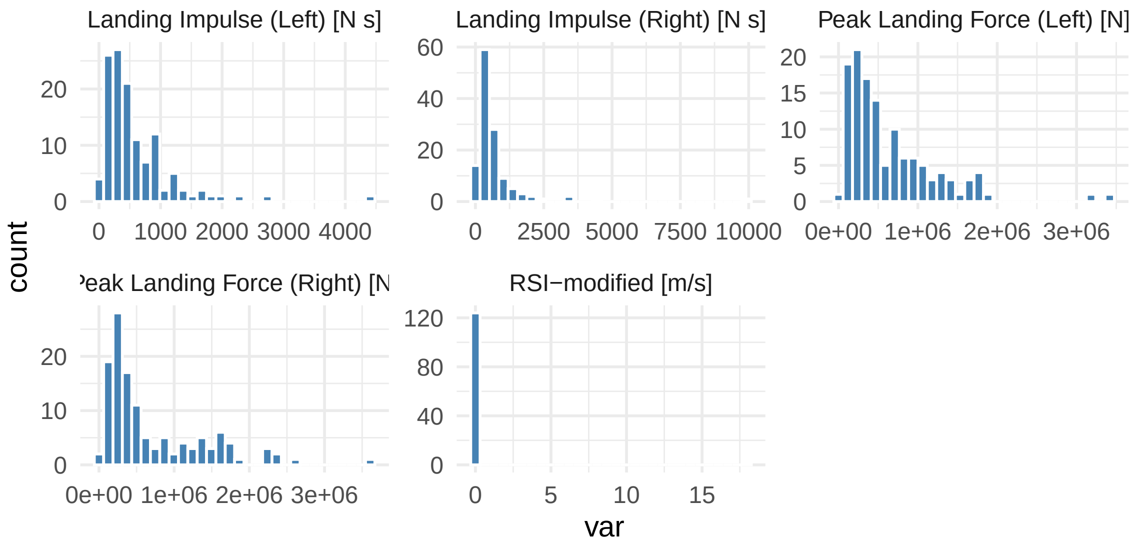
Scale−Location
√|Standardized residuals|
Fitted values
lm(‘innerRightMaxForce_Hip AD/AB − 60‘ ~ ‘Landing Impulse (Asymmetry) [% L

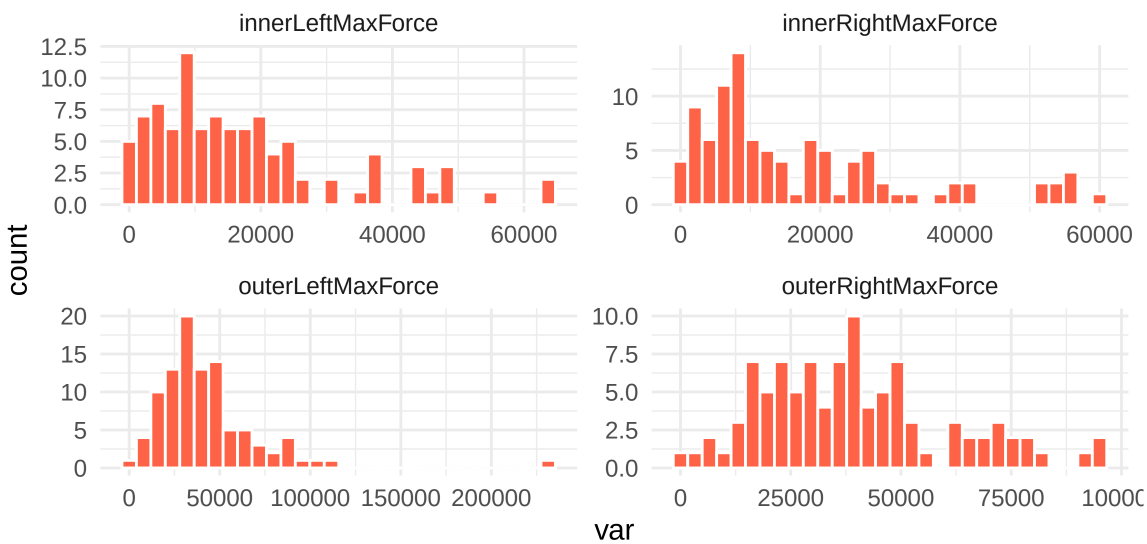


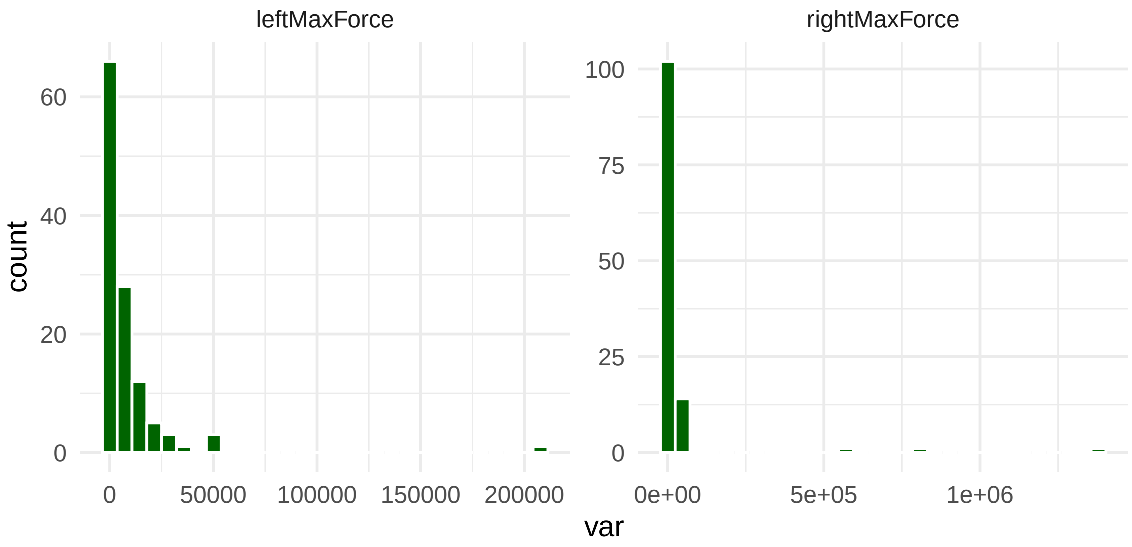
Residuals vs Leverage
Standardized residuals
Cook's distance
Leverage
lm(‘innerRightMaxForce_Hip AD/AB − 60‘ ~ ‘Landing Impulse (Asymmetry) [% L

Variance between athletes

To ensure that the aggregation of data by athlete was appropriate, we checked the variance between athletes to ensure that there was minimal variance and data was not lost.



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rightMaxForce
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100000
150000
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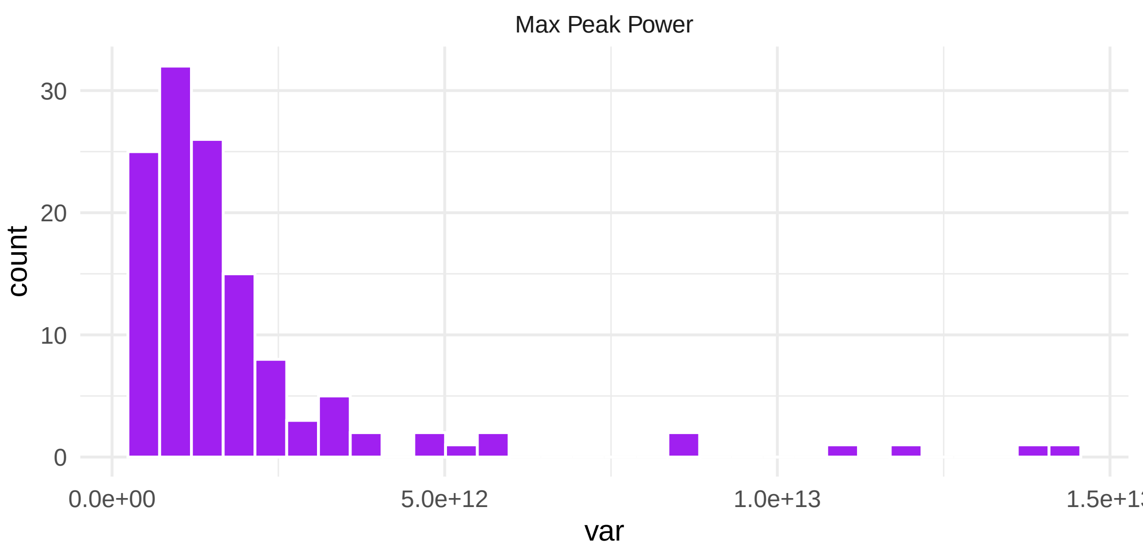
5e+05
1e+06
var


Max Peak Power
count
30
20
10
0
0.0e+00
5.0e+12
1.0e+13
var